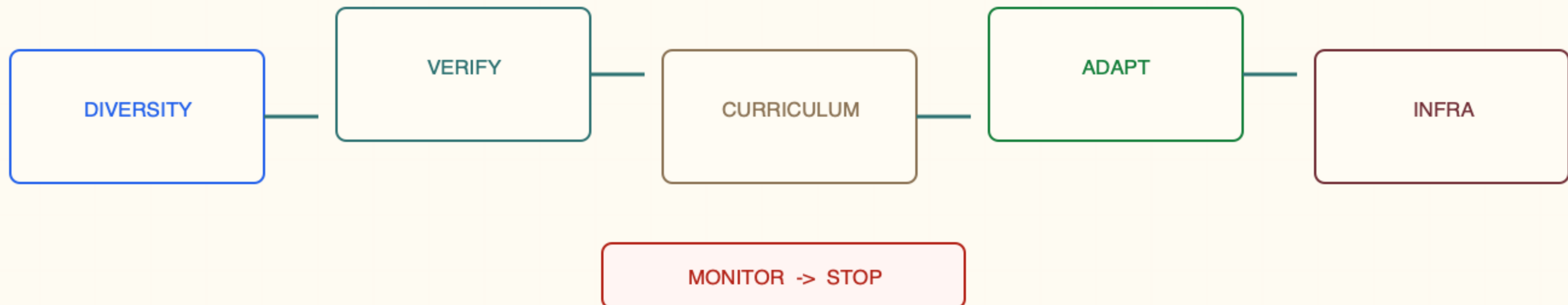


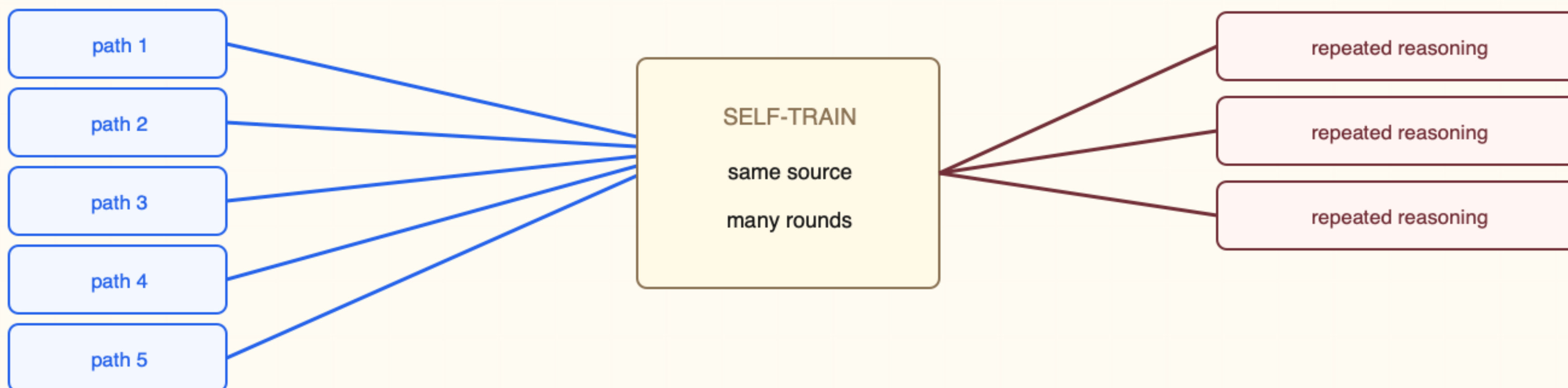
The Self-Improvement Stack

Diversity, verification, curriculum, adaptation, and infrastructure constrain one another.



Why One Model Collapses Diversity

Repeated self-training can narrow measured reasoning in the evaluated settings.

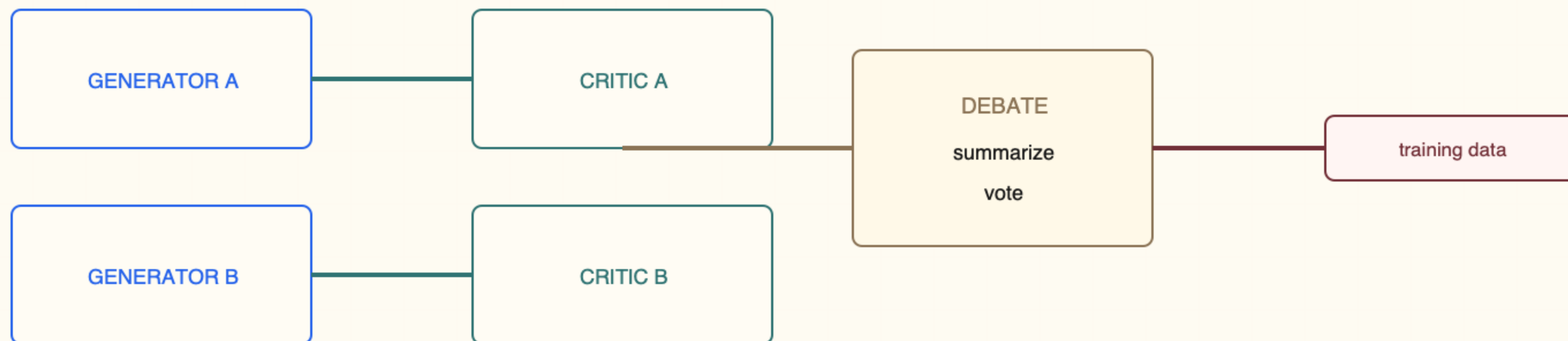


BOUNDARY

Observed in the paper's evaluated settings; not every single-model loop must collapse.

Specialize Generators and Critics

Independent role updates create a society of solution and critique strategies.

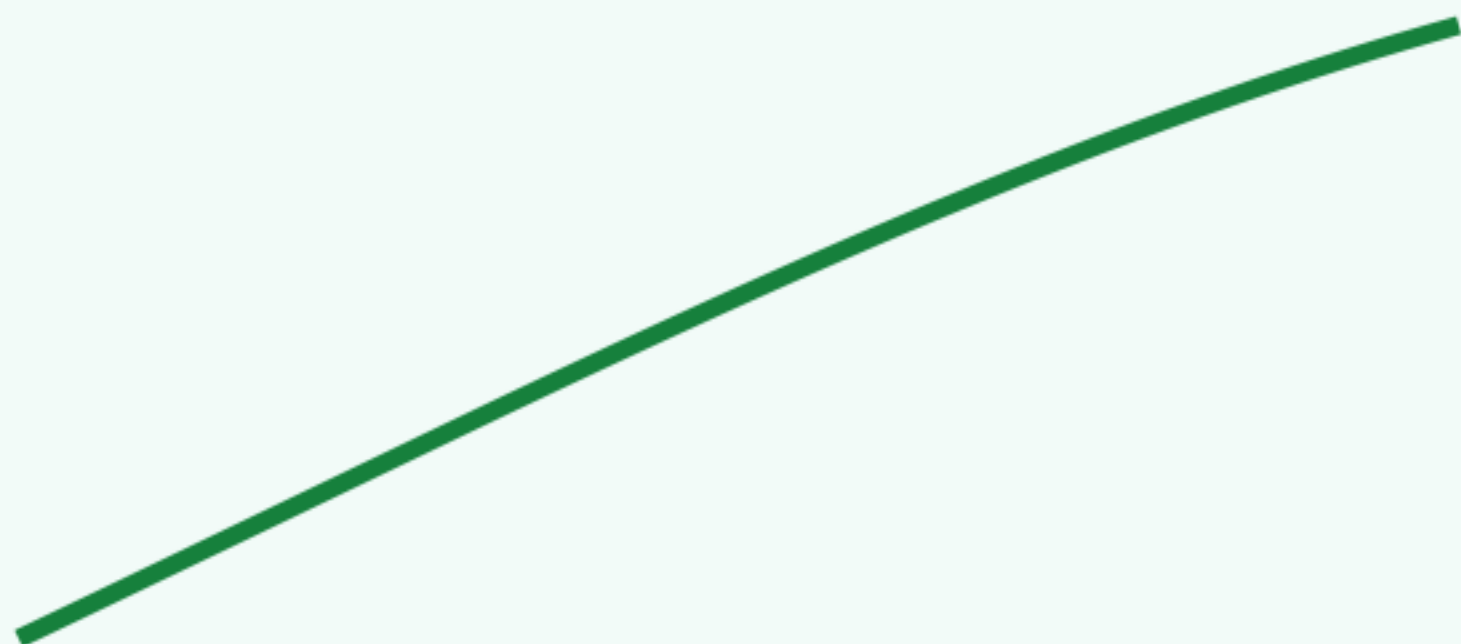
**BOUNDARY**

Specialized roles can still share correlated blind spots.

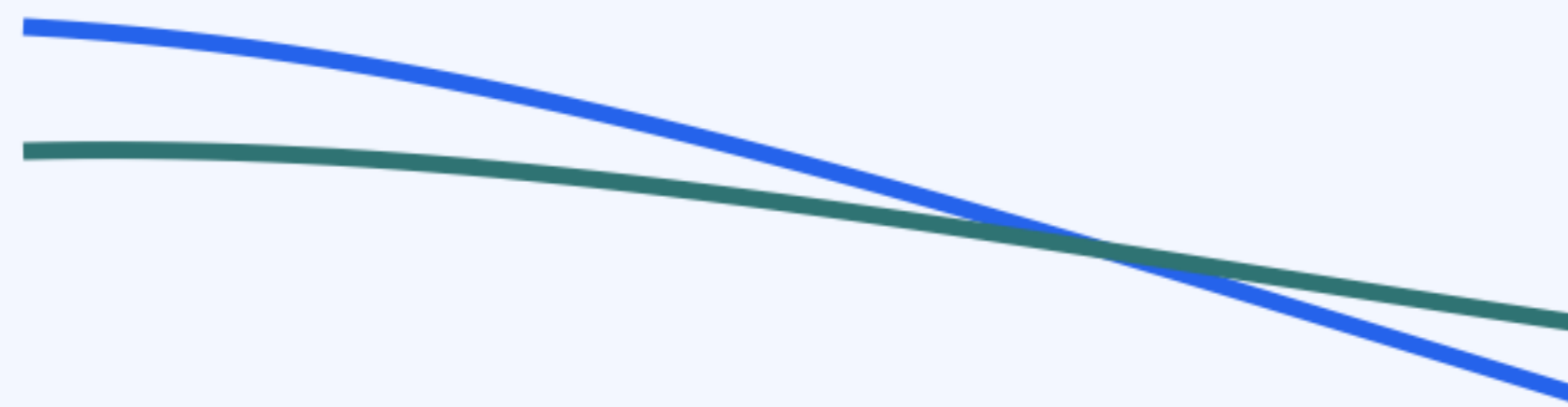
Measure Diversity, Not Just Accuracy

Useful variation supports later learning, but diversity metrics remain proxies.

ACCURACY



DIVERSITY PROXIES



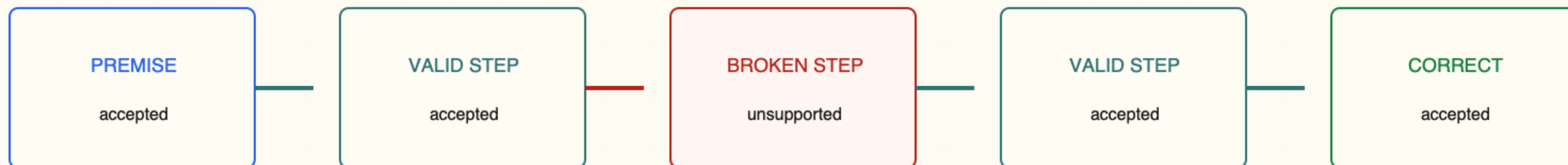
likelihood / embedding

BOUNDARY

Proxy diversity is not semantic completeness, independence, or quality.

Correct Answer, Invalid Proof

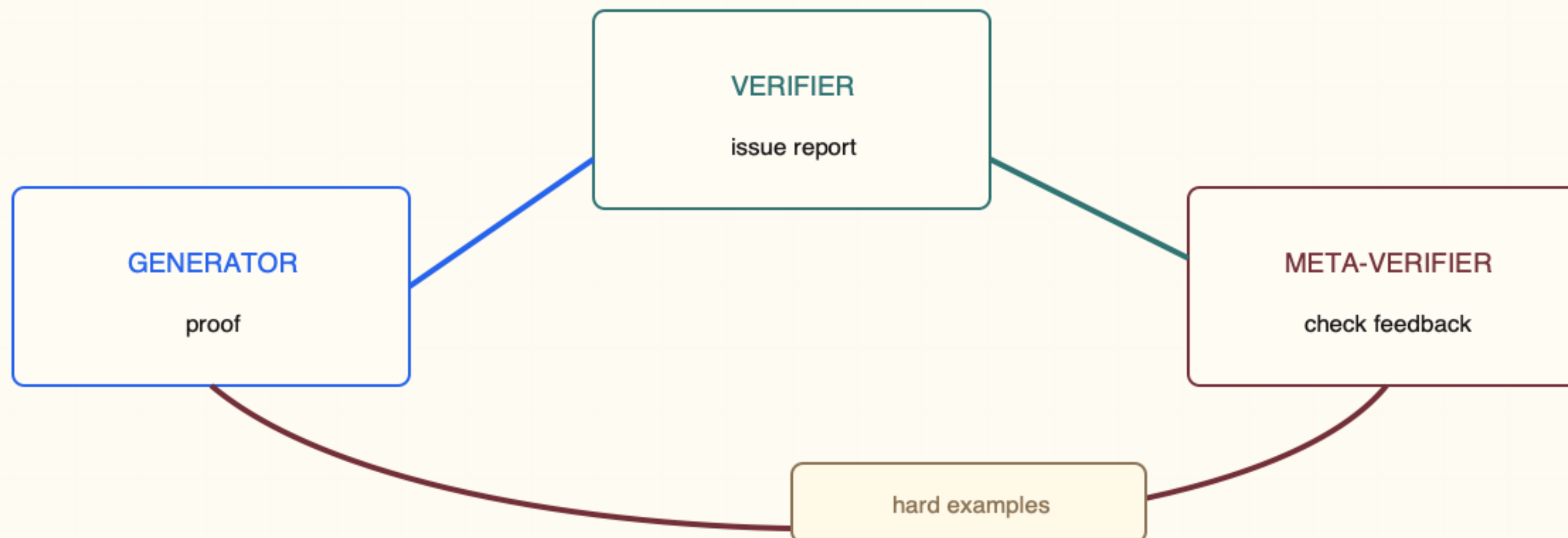
Outcome correctness cannot certify every transition in the reasoning path.



RIGHT ENDPOINT $\neq$ VALID REASONING

Verifier and Meta-Verifier

Check the proof, then check the feedback that judged the proof.

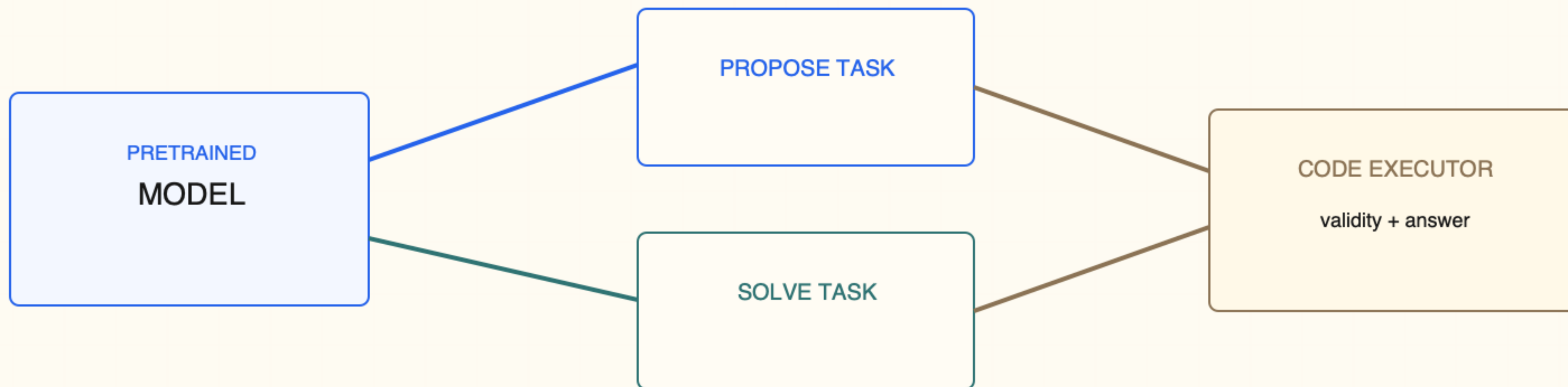


BOUNDARY

A meta-verifier moves the trust boundary; it does not create certainty.

Let the Model Propose Tasks

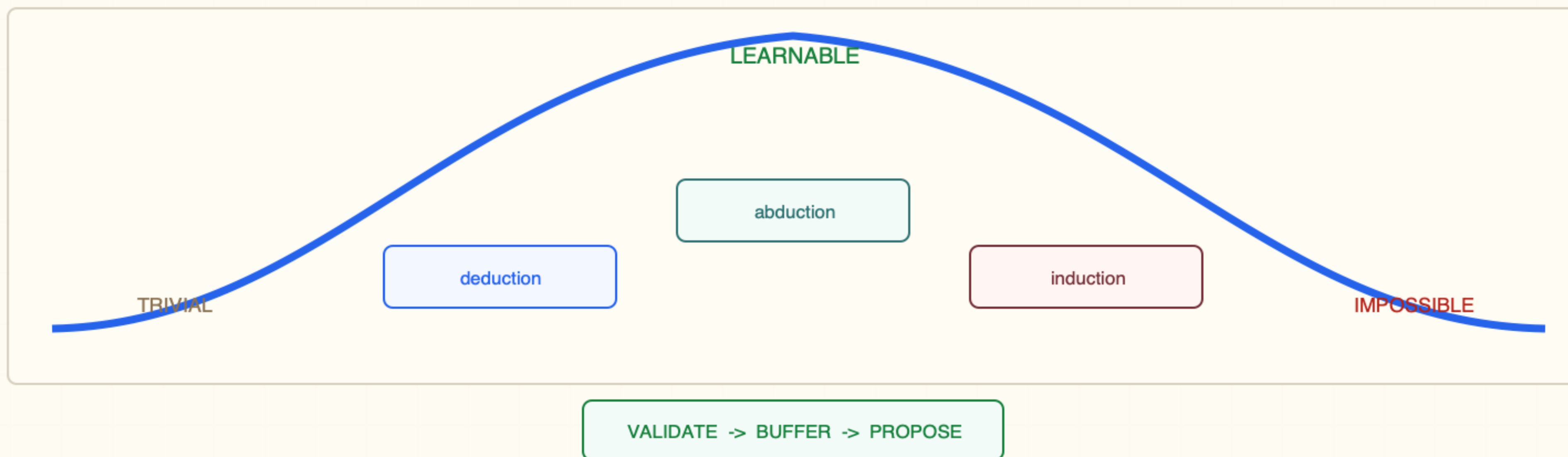
Absolute Zero joins proposer and solver without an external task dataset.

**BOUNDARY**

Zero external task data in this loop does not mean zero pretraining or structure.

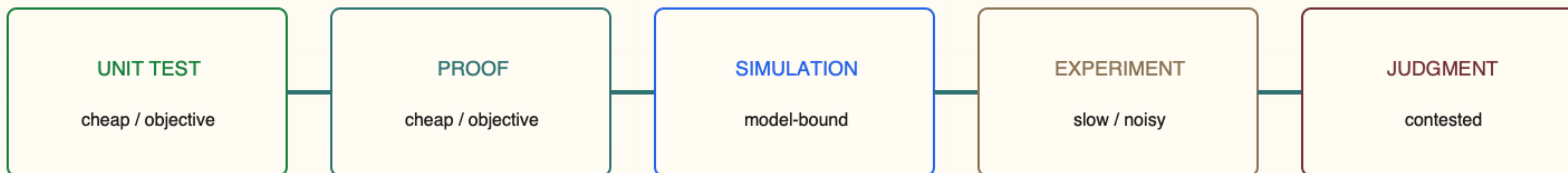
Build a Learnable Curriculum

Valid, intermediate-difficulty tasks feed an evolving buffer.



Verification Is the Boundary

Feedback becomes slower, costlier, and more contested beyond executable domains.



COST + AMBIGUITY RISE

BOUNDARY

Learned reward extends reach while increasing misspecification risk.

Intelligence per Watt

Average task accuracy divided by average power, under a bounded workload and system boundary.

INTELLIGENCE / WATT

average task accuracy

average power

CURRENT V5 SCOPE

20+ local models

8 accelerators

1M single-turn queries

REPORTED

88.7% best-of-local

5.3x historical IPW

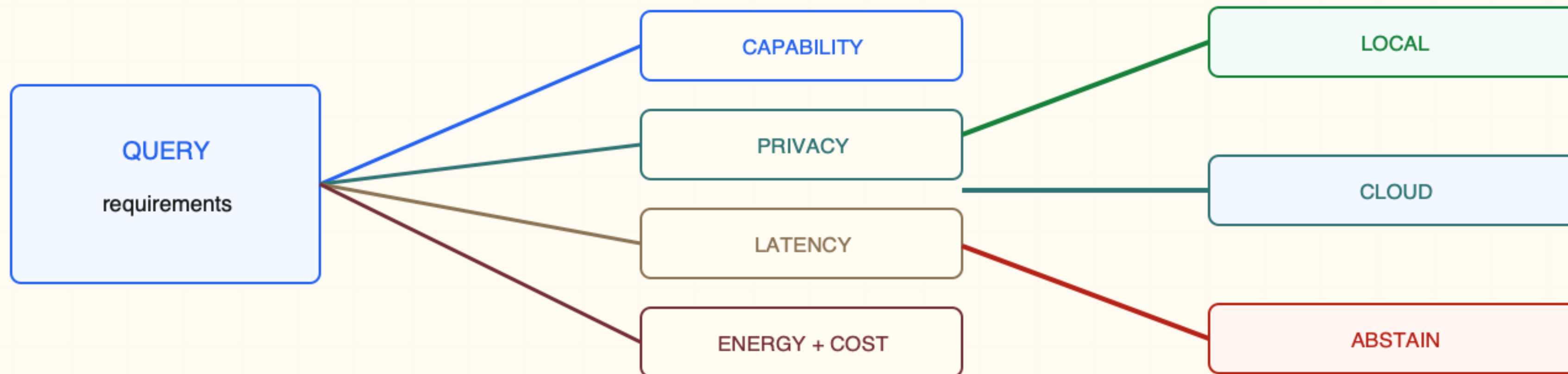
local $\geq 1.4x$ lower IPW

BOUNDARY

Versioned study results; not all requests, not a forecast, and not a universal hardware ranking.

Local, Cloud, or Hybrid?

Route by capability, privacy, latency, energy, and cost; abstain when requirements fail.



BOUNDARY

Routing is multi-objective; no path wins every workload.

The Research Agenda

Different experience; trustworthy feedback; learnable tasks; durable adaptation; efficient delivery.

